

FORTUNE MOGOENG

Senior Cloud Data Engineer | AI Engineering | FinTech | Business Intelligence Architect

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PROFESSIONAL SUMMARY

Strategic and results-driven Senior Cloud Data Engineer & FinTech BI Architect based in Cape Town, South Africa, with 8+ years of foundational experience spanning cloud data platforms, automated ETL/ELT pipelines, business intelligence, and POPIA-compliant governance. Proven track record across tier-one banking institutions (WesBank, Absa Group) and technology leaders (Embelo, ExploreAI Academy, XDS). Expert in architecting resilient batch and streaming data pipelines across Microsoft Azure (Data Factory, ADLS Gen2, Synapse, Repos) and Amazon Web Services (AWS Redshift, S3, RDS, IAM). Deep technical mastery in Python, PySpark, advanced SQL/T-SQL, Kimball dimensional data modeling, executive Power BI (DAX) models, and automated data reconciliation frameworks. Certified in Anthropic AI Agent Engineering (Claude API & Model Context Protocol) and Infrastructure as Code (Terraform).

CORE COMPETENCIES & TECHNICAL SKILLS

- **Cloud & Data Platforms:** Microsoft Azure (Azure Data Factory, ADLS Gen2, Synapse Analytics, Azure DevOps, Azure Storage), Amazon Web Services (AWS Redshift, S3, RDS, Secrets Manager, CloudWatch), Snowflake Cloud, Medallion Lakehouse Architecture.
- **Data Engineering & ETL:** Python (Pandas, PySpark, NumPy), SQL (PostgreSQL, T-SQL, MySQL, SQL Server), Apache Airflow (DAGs, Operators, Scheduling), PySpark Distributed Computing, dbt, SSIS, REST APIs, Webhooks.
- **FinTech Analytics & BI:** Microsoft Power BI (Advanced DAX, Tabular Models, Power Query M-code, RLS), MetaBase, Tableau, SSRS/SSIS, Transactional Flow Analytics, Credit Risk & Scoring Analytics, Month-End Close Reporting Automation.
- **AI & Intelligent Automation:** Anthropic Claude API Integration, Model Context Protocol (MCP Server & Tool Schemas), Claude Code, Agentic Skills Automation, RPA (UiPath, Automation Anywhere), Web Automation (Selenium).
- **Governance & POPIA Compliance:** POPIA Compliance (Protection of Personal Information Act), GDPR, Data Anonymization, PII Protection, Field-Level Encryption, Role-Based Access Control (RBAC), Automated Data Quality & Audit Reconciliation.
- **DevOps & Infrastructure:** Terraform (IaC), AWS CloudFormation, Azure Repos, GitHub Actions, Linux/Bash Scripting, Docker.

KEY TECHNICAL PROJECTS & FEATURED ARCHITECTURES

• SQL Data Engineering & Market Intelligence Pipeline (Public GitHub) [[GitHub Repo](#)]

Tech Stack: PostgreSQL, Advanced SQL, CTEs, Window Functions, EDA

- Analyzed high-demand data skills and salary distributions across tech roles using PostgreSQL CTEs, aggregations, and window functions.
- Engineered optimal skill intersection queries to identify high-ROI career paths; structured modular, version-controlled SQL scripts.

• Vehicle & Asset Finance Stock Reconciliation Pipeline

Tech Stack: Python, SQL / T-SQL, Pandas, Dimensional Modeling, Reconciliation

- Engineered high-throughput relational data transformations and reconciliation pipelines joining multi-table automotive datasets (vehicle stock, MM catalog specs, image registries, and service logs) to eliminate catalog mismatch and audit inventory in real-time.
- Automated mismatch identification between vehicle MM master catalogs and active dealer stock, creating audit-ready reporting views.

• Anthropic Claude API & MCP Autonomous Data Agent

Tech Stack: Anthropic Claude API, Model Context Protocol (MCP), Claude Code, Python

- Architected an autonomous database assistant and ETL tool orchestrator leveraging the Anthropic Claude API and Model Context Protocol (MCP).
- Implemented custom MCP servers to expose database metadata securely to LLMs, enabling natural language SQL generation and validation.

• Azure Cloud Data Lakehouse & ADF Pipeline

Tech Stack: Azure Data Factory, ADLS Gen2, Synapse Analytics, Terraform, CI/CD

- Designed scalable cloud data architectures across Microsoft Azure, orchestrating automated multi-source ingestion into medallion lakehouse storage layers with automated monitoring and zero-trust IAM.

• Power BI Financial & Credit Risk Executive Analytics

Tech Stack: Microsoft Power BI, Advanced DAX, Power Query (M), SSRS

- Developed enterprise-grade tabular data models and dynamic Power BI dashboards for senior leadership, delivering real-time visibility into credit risk exposure, loan book settlement velocity, and transactional flow bottlenecks.

PROFESSIONAL WORK EXPERIENCE

Senior Cloud Data Engineer | Embelo

City of Cape Town, South Africa (Hybrid) • Nov 2025 – Present

- Data Pipeline Architecture: Design, develop, and automate robust batch and real-time ETL/ELT data pipelines to ingest, transform, and load complex datasets from disparate enterprise sources.
- Cloud Infrastructure Management: Build and optimize scalable cloud data warehouses, data lakes, and storage solutions within enterprise cloud environments utilizing Azure Data Factory, Azure Data Lake, and Azure Storage.
- System & Query Optimization: Monitor, tune, and optimize database performance, indexing, and query execution times to ensure high availability and cost-effective cloud resource consumption.
- Data Governance & Quality: Implement data quality assurance protocols, validation checks, and data governance frameworks to maintain high standards of data integrity, security, and compliance.
- Cross-Functional Collaboration: Partner with BI engineers, data scientists, product owners, and business analysts to translate complex business requirements into scalable technical data models.
- Automation & CI/CD: Implement CI/CD pipelines and infrastructure-as-code (IaC via Terraform) to automate deployments, orchestrate workflows, and eliminate manual processing errors.

Business Intelligence Engineer | WesBank

South Africa (Hybrid / Contract) • Feb 2025 – Nov 2025

- POPIA Compliance & PII Anonymization: Designed and implemented automated data-masking and encryption protocols, ensuring full POPIA compliance during the migration of sensitive financial and customer datasets across SQL Server and AWS environments.
- Data Quality & Reconciliation Engine: Built a robust Data Quality, Validation & Reconciliation framework using Python, reducing financial-reporting discrepancies and enabling a reliable single source of truth for audits.
- Transactional Flow Optimization: Developed high-performance SQL views, stored procedures, and transactional-flow analytics to identify bottlenecks in high-volume Credit/Debit processing for vehicle and asset finance systems.
- Reporting Automation: Improved operational efficiency by automating recurring financial reporting processes using SSRS/SSIS, saving the business many hours per week and accelerating stakeholder reporting.
- Cloud Cost & Storage Efficiency: Optimised AWS S3 and Redshift usage through data tiering and ETL performance tuning, resulting in measurable reductions in monthly cloud-infrastructure costs.
- Predictive Risk Integration: Partnered with Data Science teams to embed predictive credit-scoring models into BI dashboards, offering proactive early-warning insights for credit-risk teams.
- DPLC & Version Control: Strengthened the Data Product Life Cycle (DPLC) by introducing Azure Repos version control and CI/CD practices, reducing deployment defects.

Data Engineer | Absa Group

Sandton / Remote (Contract) • Feb 2024 – Jan 2025

- Data Warehouse Modernization (Google XYZ): Architected unified enterprise data warehouse schemas for retail and corporate banking analytics by implementing columnar storage, hash-partitioning, and clustered index hierarchies, ensuring consistent sub-second query response times across multi-source financial datasets.
- Cloud Data Solutions & Scalability: Engineered cloud-native streaming and batch ingestion pipelines on AWS (Redshift, S3, RDS) to ingest transactional banking metrics, eliminating processing bottlenecks through automated cloud scaling.
- Banking API & Cross-Platform Integration: Developed and maintained secure RESTful APIs, Webhooks, and data connectors to facilitate seamless, real-time data flow between core banking ledger systems and external regulatory services.
- Automated Data Integrity & Resiliency: Built automated cross-source validation protocols and schema compatibility checks, eliminating data synchronization discrepancies and ensuring high data availability across downstream reporting layers.
- Financial Analytics & Executive Insights: Collaborated with cross-functional banking teams to design customer satisfaction and transactional behavior analytics, enabling product managers to uncover key banking usage patterns.

Data Engineer | ExploreAI Academy

Remote (Contract) • Jul 2023 – Feb 2024

- Pipeline Automation: Designed and developed robust, scalable data pipelines to streamline data processing and integrate multiple data sources using Apache Airflow, Spark, and SQL, improving processing speed and efficiency.

- Database Latency Optimization: Optimized database schemas and data models to improve query performance and reduce latency via indexing strategies, partitioning, and query optimization.
- Regulatory Compliance & Security: Enforced strict adherence to GDPR, CCPA, and industry security standards, implementing robust data encryption, anonymization techniques, and role-based access controls.
- Machine Learning Support: Preprocessed and curated high-volume feature stores, engineered input features, and validated pipeline outputs to support production machine learning models.
- Governance & Metadata Management: Established data lineage mapping, metadata management standards, and trained internal stakeholders on data stewardship best practices.

Data Analyst | Upwork

Remote (Freelance) • May 2022 – Jun 2023

- Data Accuracy & Verification: Entered and processed large volumes of operational and financial data accurately, consistently maintaining a data accuracy rate of 99% using advanced verification software.
- Discrepancy Resolution: Performed meticulous data validation processes, identifying and correcting 95% of errors found in data submissions through custom error-tracking systems.
- Compliance & Auditing: Conducted regular quality checks and compliance audits to safeguard sensitive records and adhere to evolving data governance policies.
- Workflow Automation: Collaborated to identify data capture bottlenecks, streamlining manual processes with automation scripts to reduce turnaround time.

Business Data Analyst & Data Architect | Xpert Decision Systems (XDS) Pty Ltd

Johannesburg (Remote / Full-time) • Mar 2018 – Apr 2022

- Cloud Architecture for Banking Clients: Partnered directly with tier-one banking and wealth-management clients to scope, gather, and architect custom cloud reporting configurations across AWS and Azure with zero production defect rate.
- RPA & Automation Delivery: Designed and deployed tailored Robotic Process Automation (RPA) workflows (Automation Anywhere & UiPath) to eliminate repetitive manual reporting operations.
- Requirements & Business Analysis: Translated complex stakeholder requirements into clear technical specifications, database schemas, and SSRS reports for wealth management teams.
- Data Governance & QA Protocols: Established comprehensive data governance frameworks and QA protocols, conducting regular audits to maintain high standards of data accuracy and regulatory compliance.

LICENSES & CERTIFICATIONS

- Building with the Claude API – Anthropic (Mar 2026) [ID: cf75s324orbc] [\[Verify Certificate ↗\]](#)
- Introduction to Model Context Protocol (MCP) – Anthropic (Mar 2026) [ID: dqrut2kbqp9a] [\[Verify Certificate ↗\]](#)
- Claude Code in Action – Anthropic (Mar 2026) [ID: qo63vmkhfvy6] [\[Verify Certificate ↗\]](#)
- Certificate of Completion: Introduction to Agent Skills – Anthropic (Mar 2026) [ID: n5jgwq7t8ceo] [\[Verify Certificate ↗\]](#)
- Data Engineering (NQF Level 5) – Sand Technologies (Jun 2023 – 2024) [NQF Level 5] [\[Verify Certificate ↗\]](#)
- WebServices/REST API Testing with SoapUI + Real-Time Projects – Udemy (Sep 2020) [ID: UC-22a36ffc-f90e-404b-acc3-ba945b3fe41c/] [\[Verify Certificate ↗\]](#)
- Advanced Sales Analytics for Decision Making with Power BI – Udemy (May 2020) [ID: UC-e04caae3-f9b0-42fe-b56e-3788ec100f66/] [\[Verify Certificate ↗\]](#)
- Microsoft Power BI - Up and Running with Power BI Desktop – Udemy (May 2020) [\[Verify Certificate ↗\]](#)
- Microsoft Power BI - Publishing to Power BI Services – Udemy (May 2020) [\[Verify Certificate ↗\]](#)
- Power BI - Data Analytics Essentials with Power BI – Udemy (May 2020) [\[Verify Certificate ↗\]](#)
- Complete UiPath RPA Developer Course: Build 7 Robots – Udemy (May 2020) [\[Verify Certificate ↗\]](#)
- SQL Boot Camp & Database Performance – Udemy (Dec 2019) [ID: UC-S999T2EZ] [\[Verify Certificate ↗\]](#)
- Microsoft Excel - Excel Power Query, Power Pivot & DAX – Udemy (Apr 2020) [ID: UC-c7cc539e-2e5c-4f17-b8f9-b44ac43bf9a6/] [\[Verify Certificate ↗\]](#)
- Microsoft Excel - Data Visualization, Charts & Graphs – Udemy (Apr 2020) [\[Verify Certificate ↗\]](#)
- Microsoft Excel - Advanced Formulas & Functions – Udemy (Apr 2020) [\[Verify Certificate ↗\]](#)

- Excel PRO TIPS: 75+ Tips to go from Excel Beginner to Pro – Udemy (Apr 2020) [\[Verify Certificate ↗\]](#)

EDUCATION & ACADEMIC QUALIFICATIONS

Bachelor of Information Systems (Information Technology) – Independent Institute of Education (IIE) (Graduated 2017)

Core Focus: Database Management Systems, Python Programming, Business Systems Analysis, SQL, Data Analytics, Data Visualization.

Data Engineering Certificate (NQF Level 5) – Sand Technologies (Graduated 2024)

Specialization: SQL Server and Database Management, Python, Data Warehousing, ETL Processes, Power BI, Big Data (Hadoop, Spark), Cloud Computing.